

10 (**long**)

(a) Show that the Laplace Transform of $f(t)e^{-at}$ is $F(s+a)$, where $F(s)$ is the Laplace Transform of $f(t)$ and where a is a constant. [3]

(b) Using Laplace Transforms, and no other method, solve the differential equation

$$\frac{dy}{dt} + y = e^{-t} \sin t$$

subject to the boundary condition $y(0) = 1$. [11]

(c) Using the relationship that relates the Laplace Transform of the convolution of two functions

$$f * g = \int_{\tau=0}^t f(\tau)g(t-\tau)d\tau$$

to those of the individual functions, find the inverse transform of

$$\frac{1}{(s^2 + 1)^2} \quad [11]$$

(d) Hence find the solution of the differential equation

$$\frac{d^2y}{dt^2} + 2\frac{dy}{dt} + 2y = e^{-t} \sin t$$

subject to the boundary conditions $y = 0$ and $dy/dt = 0$ at $t = 0$. [5]